FileMind

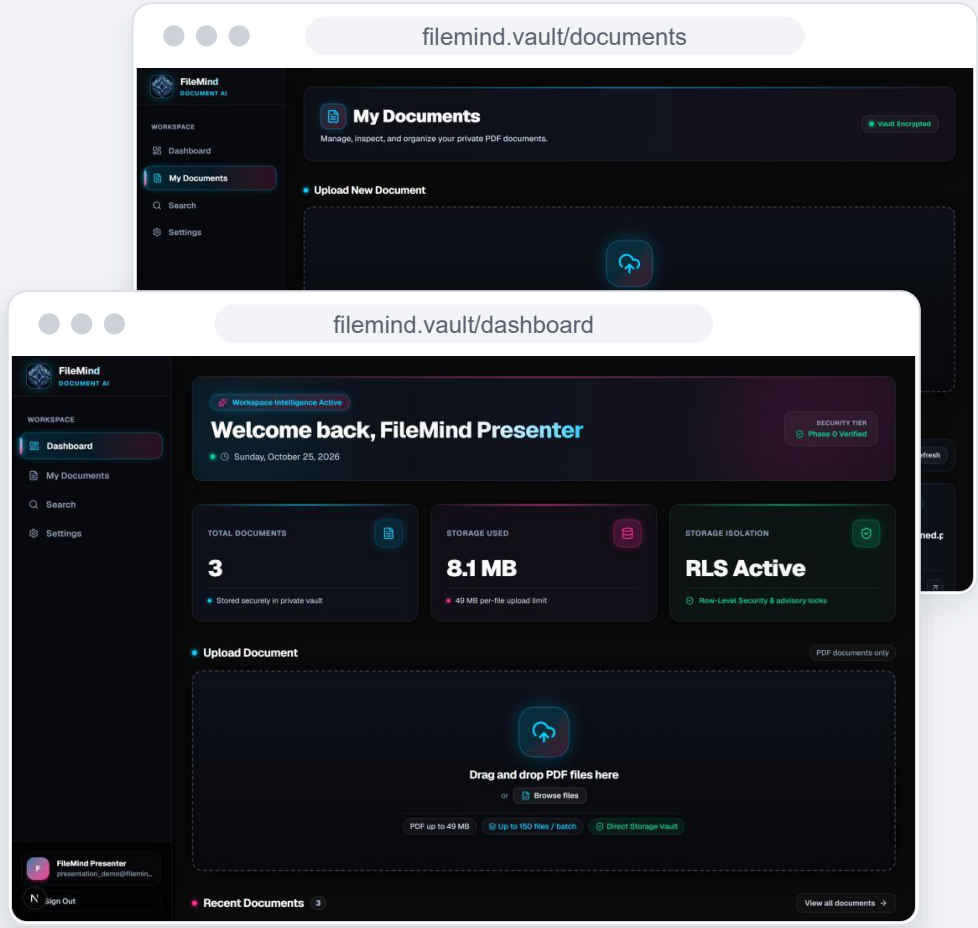
An AI-powered PDF vault that finds answers, not just files.

Upload a batch of PDFs: invoices, research papers, contracts, even scans. FileMind reads every page with multimodal AI, pulls out structured data and makes the whole vault searchable by meaning, with each user's files isolated at the database level.

150
PDFs per batch

49 MiB
max file size

< 200 ms
vector retrieval



Jitendar (Team Lead) · Kaushal Verma · Vishal
Roll: 25001011024 (IT) · 25001011027 (IT) · 25001015070 (ENC)
J.C. Bose University of Science and Technology, YMCA

Search can’t see inside your PDFs

File systems treat PDFs as opaque blobs, so the knowledge inside them is fragmented and hard to reach.

80%

of enterprise and academic knowledge is trapped in unstructured documents.

Source: IDC / Gartner

2.5 hrs

a day spent hunting for buried information: nearly 20% of the working week.

Keyword-blind storage

OS and cloud search only index file names. Scanned invoices, image-based PDFs and multi-column papers stay invisible to every query.

Manual fact extraction

Finding an invoice total, a renewal date or an indemnification clause means opening the file, skimming it and copy-pasting by hand.

Tools that break at scale

Desktop OCR tools crash on batches of 50+ files, and consumer cloud drives lack row-level access control, so private files can leak.

From passive folders to a searchable vault

FileMind replaces manual upkeep with document understanding and instant retrieval.

TRADITIONAL PDF STORAGE

Passive folders, manual upkeep

- **Exact filename search**
  Fails unless you recall ‘report_v2_final.pdf’.
- **Blind to scanned text**
  Image PDFs and scans can’t be indexed.
- **Manual data entry**
  Totals, vendors and dates typed by hand.
- **No tenant isolation**
  Shared folders with no row-level security.

FILEMIND AI VAULT

Reads documents, finds answers

- **Semantic concept search**
  Ask for ‘cloud server bill’, no keyword needed.
- **Multimodal understanding**
  Gemini extracts category, tags and page text.
- **Structured JSON output**
  Invoice totals, dates and parties as clean JSON.
- **Database-enforced isolation**
  PostgreSQL RLS plus advisory locks.

Four capabilities, verified by tests

Every feature below runs in the codebase today and is covered by an automated test suite.

BULK INGESTION

150 files

Sliding-window batch concurrency

Up to 150 PDFs per batch, 49 MiB each. Client-side windowing keeps at most 4 uploads and 2 background AI workers active, with per-file error isolation.

MULTIMODAL INTELLIGENCE

Gemini 3.6

Structured schema extraction

Classifies each file (invoice, contract, report), extracts topic tags and key-value fields, and blocks prompt injection with strict system instructions.

SEARCH & RETRIEVAL

< 200 ms

Hybrid semantic vector search

Cosine similarity over 768-dim embeddings plus a 500-entry in-memory LRU cache. Queries are normalised and duplicate in-flight requests collapsed.

ENTERPRISE SECURITY

100% RLS

Multi-tenant lease isolation

Row-Level Security blocks cross-tenant reads. Workers claim jobs with SKIP LOCKED and UUID fencing tokens: no double-processing, no stale writes.

From drag-and-drop to searchable in five steps

The lifecycle of every file, from browser upload to an indexed, searchable document.

1

INGEST

Client batch upload

The browser checks the 49 MiB limit, then a sliding-window queue streams files straight to the private bucket, with no server bottleneck.

2

VERIFY

Atomic quota check

A stored procedure confirms the object exists and takes `pg_advisory_xact_lock` to enforce the 2 GB per-user quota.

3

LOCK

Queue lease claim

A worker claims the next job with `FOR UPDATE SKIP LOCKED` and gets a UUID fencing token, so no file is processed twice.

4

UNDERSTAND

Multimodal extraction

The PDF goes to the Gemini Files API. A model cascade (3.6-flash → 3.8-flash → preview) with backoff extracts category, tags and pages.

5

INDEX

Vector index commit

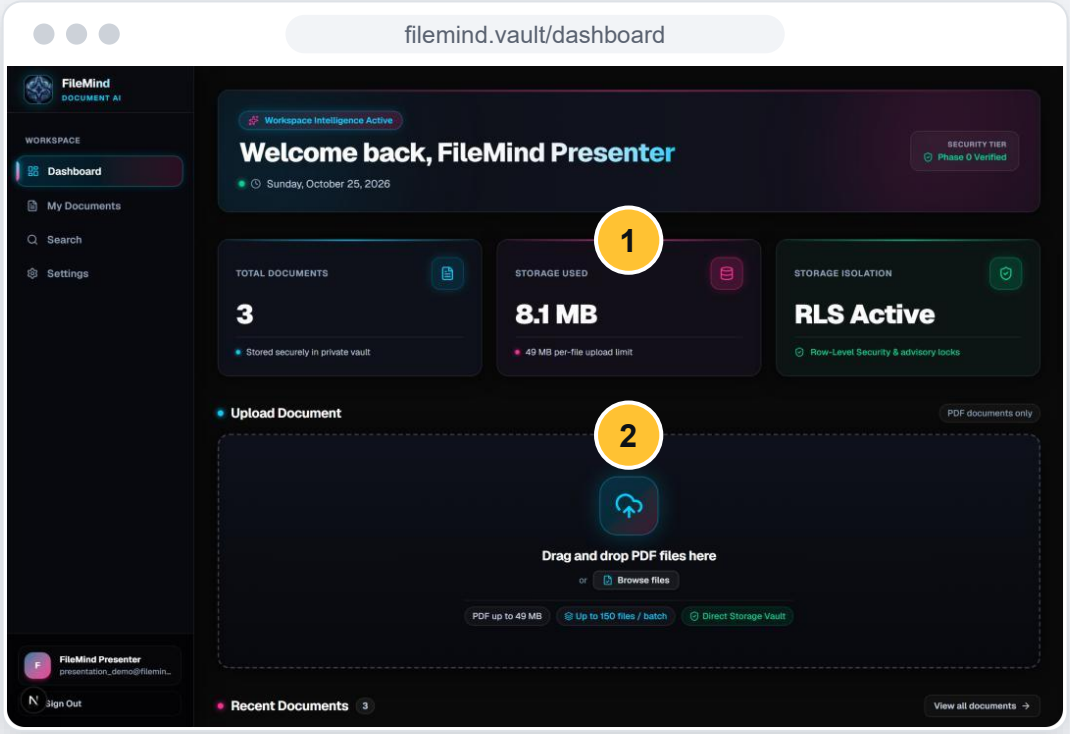
The finalize RPC validates the lease token, commits metadata atomically and schedules 768-dim embeddings.

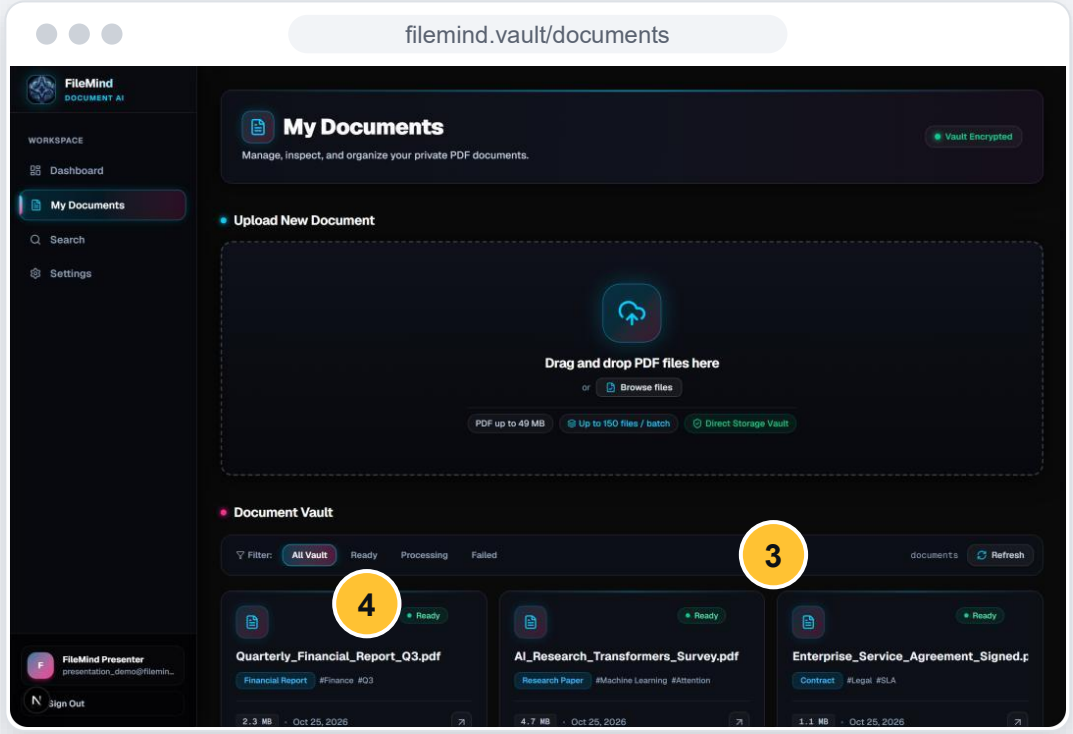
BUILT TO FAIL SAFELY

- Per-file error isolation
- Exponential backoff with jitter
- Background jobs via pg_cron

Built and running today

The working app: a dark-mode interface built with Next.js 16, Tailwind CSS v4 and Lucide icons.





1. Live storage quota metrics
2. 150-file batch drop zone
3. Ready / processing / failed
4. Gemini categories & tags

A modern, type-safe stack

Production-grade frameworks across the interface, the data layer and the AI pipeline.

FRONTEND & APPLICATION

Server-rendered, streaming UI

Next.js 16.3 (App Router)

Turbopack, Server Components and `after()` for background work.

React 19.2

Streaming transitions and optimistic UI updates.

Tailwind CSS v4

High-contrast dark theme with no CSS-in-JS runtime.

BACKEND, DATA & SECURITY

Isolation and atomic locking

Supabase PostgreSQL 15+

Row-Level Security for strict multi-tenant vault isolation.

PL/pgSQL stored procedures

Atomic RPCs:

```
create_document_with_quota
claim_next_pending_document
finalize_document_processing
```

Supabase Storage (S3)

Private bucket with authenticated, presigned download URLs.

MULTIMODAL AI & ACCELERATION

Vision AI and vector caching

Gemini API (@google/genai)

Direct PDF ingestion via the Files API, JSON-schema output and prompt-injection defence.

Vector embeddings (768-dim)

Hybrid semantic search with a 500-entry LRU query-embedding cache.

pdf-lib

Lossless stream compression that safely skips signed or encrypted PDFs.

Four layers, with security in the database

Clear separation between the UI, the serverless gateway, the data tier and the AI engine.

01 BROWSER

Client presentation

- App Router UI
- Dashboard
- Documents vault
- PDF.js viewer
- Semantic search explorer
- Batch queue: 4 concurrent uploads, per-file retry

↓

02 NEXT.JS 16

Gateway & serverless

- Server Actions
- `/api/documents`
- `/api/search`
- `/api/cron/process-queue`
- after() execution budgets
- Supabase Auth JWT sessions

↓

03 SUPABASE

Database & storage

- PostgreSQL + Row-Level Security
- Atomic PL/pgSQL RPCs
- Transactional advisory locks
- `private bucket: documents`

↓

04 GEMINI

AI & acceleration

- Gemini Files API
- Structured JSON extraction
- Category & key-value metadata
- 768-dim embeddings
- 500-entry LRU cache
- pdf-lib stream optimiser

How FileMind compares

Against local OS folders, generic cloud drives and OCR utilities.

Capability	Local OS folders	Generic cloud drive	FileMind AI vault
Search method	Exact filename match only	Keyword match on text-based PDFs	**Semantic vector retrieval (768-dim embeddings)**
Document understanding	None; manual reading	Basic text OCR	**Structured JSON: category, tags, key-values**
Scanned & image PDFs	Opaque and unsearchable	Garbled multi-column text	**Read page by page with multimodal vision AI**
Bulk processing	OS file copy, no validation	Slow sequential sync	**150-file sliding window, 4 concurrent streams**
Security & isolation	Local OS user permissions	Coarse folder sharing links	**Row-Level Security and fencing leases**

Hard problems we solved along the way

Real failures we hit during development, and what we built to fix each one.

CHALLENGE	WHAT WENT WRONG	WHAT WE BUILT
Gemini 503 spikes at peak hours	Google's standard tier returned 503 UNAVAILABLE at peak times, failing queue jobs and stranding files.	3-tier fallback with backoff and jitter: `gemini-3.6-flash` → `gemini-3.8-flash` → `gemini-3-flash-preview`
Serverless limits on 150-file batches	Processing 150 PDFs inside a 60-second serverless budget got functions killed mid-batch.	Sliding-window batching (max 4 uploads, 2 AI workers), Next.js `after()` execution and a `pg_cron` background consumer.
Race conditions under concurrent uploads	Simultaneous confirmations created duplicate rows and let users slip past their storage quota.	Transactional `pg_advisory_xact_lock(user_id)` plus atomic SKIP LOCKED queue claims with UUID fencing leases.
Optimising PDFs without corrupting them	Compressing signed or encrypted PDFs breaks their signatures and can leave files unreadable.	Pre-checks detect `/ByteRange` signatures and `/Encrypt` dictionaries and skip those files; everything else is optimised.

Roadmap: from vault to knowledge platform

What we plan to build next to make FileMind enterprise-ready.

PHASE 01 · LIVE

AI document vault

Batch ingestion, Gemini extraction, semantic search and RLS isolation: everything shown in this deck.

PHASE 02

Cross-document synthesis & multi-PDF Q&A

Ask one question across the whole vault, e.g. ‘Compare revenue growth and profit margins across the Q1, Q2 and Q3 reports’, and get a comparative summary.

PHASE 03

Visual citations & bounding-box viewer

Highlight the exact sentence and bounding box on the original PDF page for every generated answer, so each citation can be verified.

PHASE 04

Local & air-gapped LLM execution

Ollama and vLLM integration for fully offline inference in sensitive legal, defence and HIPAA-regulated medical environments.

PHASE 05

Team workspaces & role-based access

Admin, Contributor and Read-only Reviewer roles, with cryptographic audit logs of document access and metadata changes.

Live demo: what you'll see

We'll run the app live with a real batch of invoices, research papers and legal agreements.

1

Live ingestion walkthrough

Drag a mixed batch of test invoices, papers and contracts into the running web app and watch per-file progress.

2

Real-time multimodal extraction

See Gemini classify each document, extract structured key-values and index pages on the fly.

3

Semantic search & snippet matching

Search by concept in plain language and get sub-200 ms vector results with highlighted snippets.

TRY IT YOURSELF

Resources

Source code

`github.com/jitendarswami63-ai/file-mind`

Live demo

`filemind.vercel.app`

Test documents

Invoices, research papers and legal agreements

Stop searching for documents. Start finding answers.

Problem solved

Passive, unsearchable folders become an active AI document vault.

Production quality

Next.js 16, Supabase PostgreSQL and Gemini, backed by test suites.

Resilient at scale

Tested on 150-file batches with automated recovery and advisory locks.

Privacy by design

Row-Level Security keeps every user's vault strictly isolated.

Thank you. Questions welcome.

Jitendar (Team Lead) · Kaushal Verma · Vishal
Roll: 25001011024 (IT) · 25001011027 (IT) · 25001015070 (ENC)
J.C. Bose University of Science and Technology, YMCA